

R-Code : Case 1

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1. Package Installation (run once)

```
packages <- c(
  "tidyverse",
  "caret",
  "pROC",
  "ROCR",
  "glmnet",
  "randomForest",
  "xgboost",
  "vroom",
  "data.table",
  "janitor"
)
```

2. Setup

```
set.seed(202031137)

library(tidyverse)
library(caret)
library(pROC)
library(janitor)
```

3. Load Data

```
df <- read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv", na = c("", "NA", " "))

glimpse(df)

## Rows: 7,043
## Columns: 21
## $ customerID      <chr> "7590-VHVEG", "5575-GNVDE", "3668-QPYBK", "7795-CFOCW~
## $ gender           <chr> "Female", "Male", "Male", "Male", "Female", "Female",~
## $ SeniorCitizen    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ Partner          <chr> "Yes", "No", "No", "No", "No", "No", "No", "No", "Yes~
```

```
## $ Dependents      <chr> "No", "No", "No", "No", "No", "No", "Yes", "No", "No"~
## $ tenure          <dbl> 1, 34, 2, 45, 2, 8, 22, 10, 28, 62, 13, 16, 58, 49, 2~
## $ PhoneService    <chr> "No", "Yes", "Yes", "No", "Yes", "Yes", "Yes", "No", ~
## $ MultipleLines    <chr> "No phone service", "No", "No", "No phone service", "~
## $ InternetService  <chr> "DSL", "DSL", "DSL", "DSL", "Fiber optic", "Fiber opt~
## $ OnlineSecurity   <chr> "No", "Yes", "Yes", "Yes", "No", "No", "No", "Yes", "~
## $ OnlineBackup     <chr> "Yes", "No", "Yes", "No", "No", "No", "Yes", "No", "N~
## $ DeviceProtection <chr> "No", "Yes", "No", "Yes", "No", "Yes", "No", "No", "Y~
## $ TechSupport      <chr> "No", "No", "No", "Yes", "No", "No", "No", "No", "Yes~
## $ StreamingTV       <chr> "No", "No", "No", "No", "No", "Yes", "Yes", "No", "Ye~
## $ StreamingMovies  <chr> "No", "No", "No", "No", "No", "Yes", "No", "No", "Yes~
## $ Contract         <chr> "Month-to-month", "One year", "Month-to-month", "One ~
## $ PaperlessBilling <chr> "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes", "No", ~
## $ PaymentMethod    <chr> "Electronic check", "Mailed check", "Mailed check", "~
## $ MonthlyCharges   <dbl> 29.85, 56.95, 53.85, 42.30, 70.70, 99.65, 89.10, 29.7~
## $ TotalCharges     <dbl> 29.85, 1889.50, 108.15, 1840.75, 151.65, 820.50, 1949~
## $ Churn            <chr> "No", "No", "Yes", "No", "Yes", "Yes", "No", "No", "Y~
```

```
summary(df)
```

```
##      customerID      gender      SeniorCitizen      Partner
## Length:7043      Length:7043      Min.      :0.0000      Length:7043
## Class :character      Class :character      1st Qu.:0.0000      Class :character
## Mode  :character      Mode  :character      Median :0.0000      Mode  :character
##                                     Mean      :0.1621
##                                     3rd Qu.:0.0000
##                                     Max.      :1.0000
##
##      Dependents      tenure      PhoneService      MultipleLines
## Length:7043      Min.      : 0.00      Length:7043      Length:7043
## Class :character      1st Qu.: 9.00      Class :character      Class :character
## Mode  :character      Median :29.00      Mode  :character      Mode  :character
##                                     Mean      :32.37
##                                     3rd Qu.:55.00
##                                     Max.      :72.00
##
##      InternetService      OnlineSecurity      OnlineBackup      DeviceProtection
## Length:7043      Length:7043      Length:7043      Length:7043
## Class :character      Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character      Mode  :character
##
##
##
##      TechSupport      StreamingTV      StreamingMovies      Contract
## Length:7043      Length:7043      Length:7043      Length:7043
## Class :character      Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character      Mode  :character
##
##
##
##      PaperlessBilling      PaymentMethod      MonthlyCharges      TotalCharges
## Length:7043      Length:7043      Min.      : 18.25      Min.      : 18.8
```

```
## Class :character   Class :character   1st Qu.: 35.50   1st Qu.: 401.4
## Mode :character   Mode :character   Median : 70.35   Median :1397.5
##                                     Mean  : 64.76   Mean  :2283.3
##                                     3rd Qu.: 89.85   3rd Qu.:3794.7
##                                     Max.   :118.75   Max.   :8684.8
##                                     NA's    :11
##      Churn
## Length:7043
## Class :character
## Mode :character
##
##
##
##
```

```
tabyl(df$Churn)
```

```
## df$Churn    n    percent
##      No 5174 0.7346301
##      Yes 1869 0.2653699
```

4. Fix Data Types

```
df <- df %>%
  mutate(TotalCharges = as.numeric(TotalCharges))

sum(is.na(df$TotalCharges))
```

```
## [1] 11
```

5. Missing Value Handling

```
df <- df %>%
  mutate(TotalCharges = as.numeric(TotalCharges)) %>%
  mutate(TotalCharges = if_else(
    is.na(TotalCharges),
    median(TotalCharges, na.rm = TRUE),
    TotalCharges
  ))
```

6. Data Cleaning & Recoding

```
df <- df %>%
  mutate(
    across(where(is.character), as.factor),
    SeniorCitizen = factor(SeniorCitizen, levels = c(0,1), labels = c("No", "Yes")),
  )
```

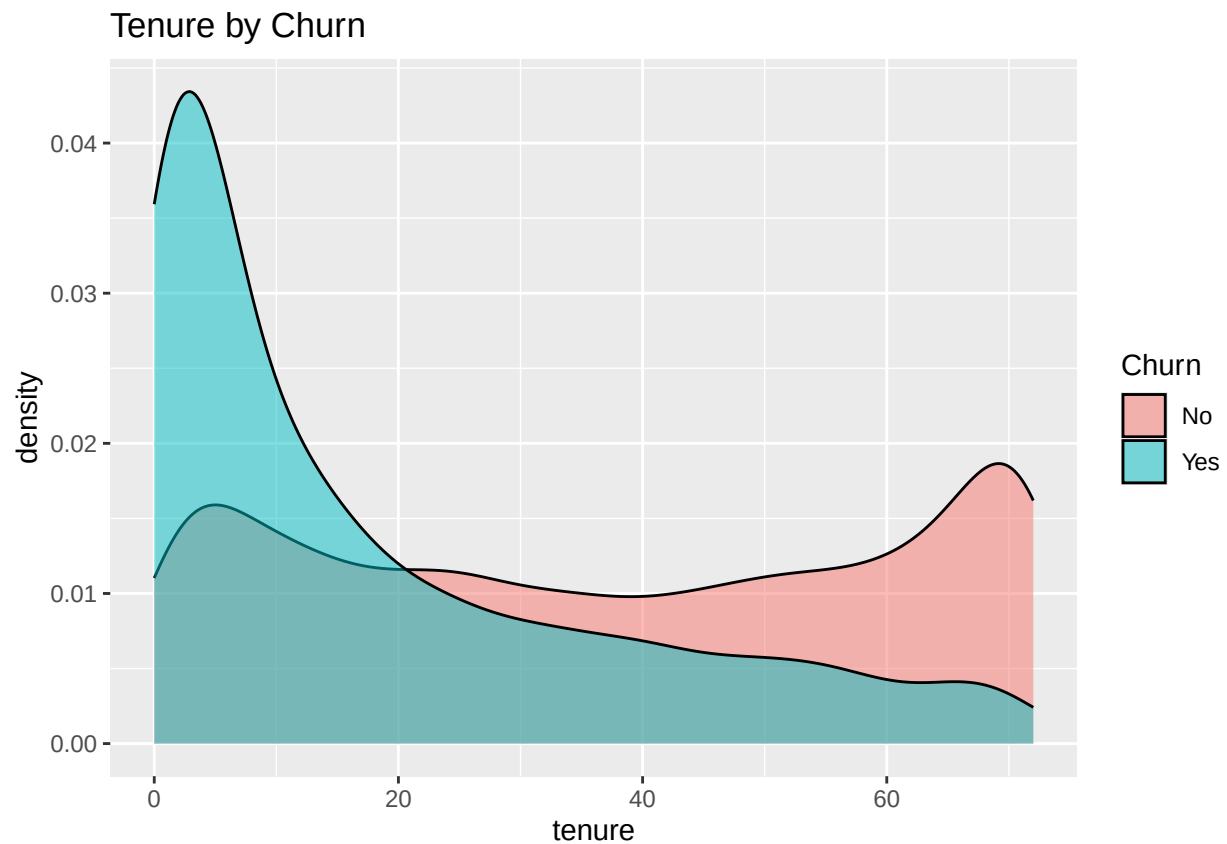
```
Churn = factor(Churn, levels = c("No", "Yes"))
) %>%
select(-customerID)
```

7. Exploratory Data Analysis

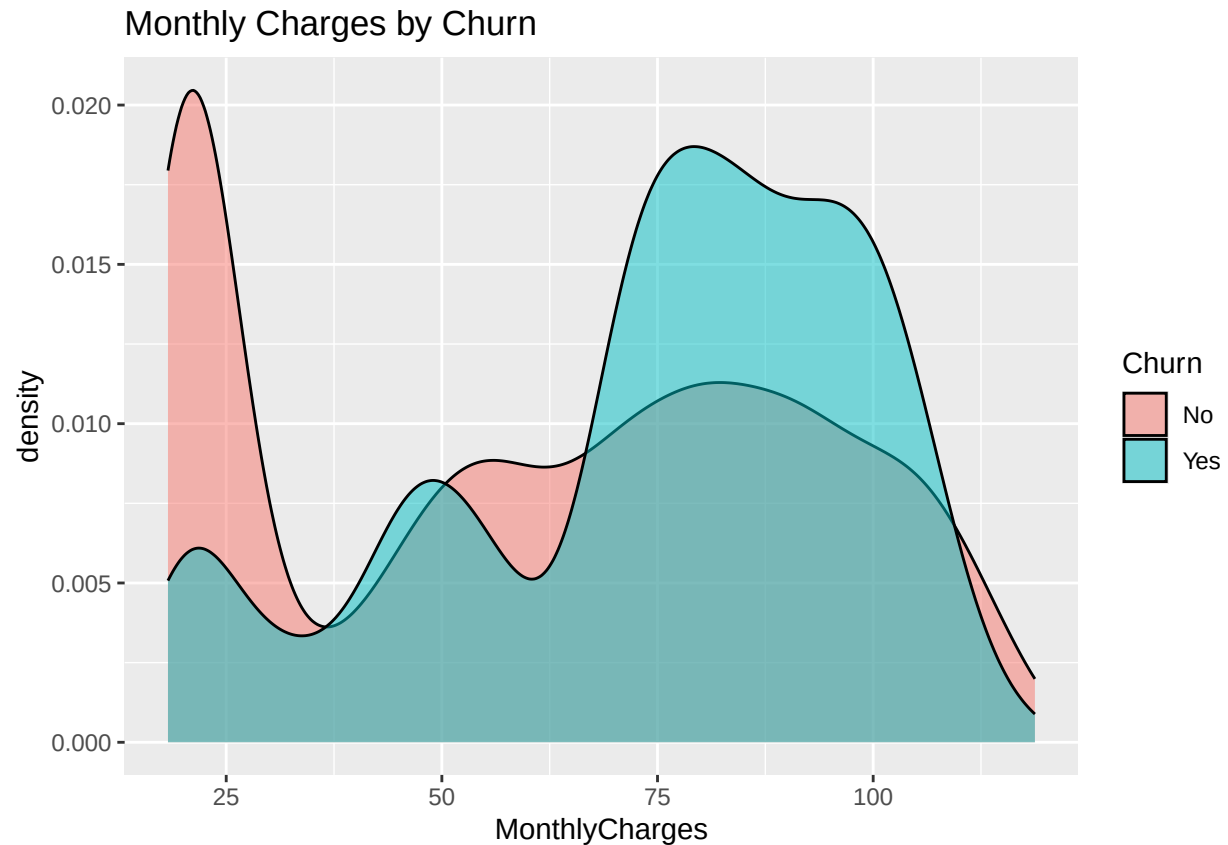
```
prop.table(table(df$Churn))
```

```
##
##          No          Yes
## 0.7346301 0.2653699
```

```
ggplot(df, aes(x = tenure, fill = Churn)) +
  geom_density(alpha = 0.5) +
  labs(title = "Tenure by Churn")
```



```
ggplot(df, aes(x = MonthlyCharges, fill = Churn)) +
  geom_density(alpha = 0.5) +
  labs(title = "Monthly Charges by Churn")
```



8. Feature Engineering

```
df <- df %>%
  mutate(
    tenure_group = case_when(
      tenure <= 12 ~ "0-12",
      tenure <= 24 ~ "13-24",
      tenure <= 48 ~ "25-48",
      tenure <= 60 ~ "49-60",
      TRUE ~ "61+"
    ),
    tenure_group = factor(
      tenure_group,
      levels = c("0-12", "13-24", "25-48", "49-60", "61+")
    )
  )
```

9. Train-Test Split

```
train_index <- createDataPartition(df$Churn, p = 0.75, list = FALSE)
train_df <- df[train_index, ]
test_df <- df[-train_index, ]
```

10. Dummy Encoding (Optional)

```
dummies <- dummyVars(Churn ~ ., data = train_df, fullRank = TRUE)

x_train <- predict(dummies, train_df) %>% as.data.frame()
x_test  <- predict(dummies, test_df) %>% as.data.frame()

y_train <- train_df$Churn
y_test  <- test_df$Churn
```

11. Logistic Regression Model

```
glm_fit <- glm(
  Churn ~ tenure + MonthlyCharges + TotalCharges +
    Contract + InternetService + tenure_group,
  data = train_df,
  family = binomial(link = "logit")
)

summary(glm_fit)
```

```
##
## Call:
## glm(formula = Churn ~ tenure + MonthlyCharges + TotalCharges +
##      Contract + InternetService + tenure_group, family = binomial(link = "logit"),
##      data = train_df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.1118603   0.2250487  -0.497   0.6192
## tenure        -0.0542342   0.0107506  -5.045 4.54e-07 ***
## MonthlyCharges  0.0009615   0.0041036   0.234   0.8147
## TotalCharges    0.0001601   0.0000868   1.845   0.0651 .
## ContractOne year -0.8349944   0.1217922  -6.856 7.09e-12 ***
## ContractTwo year -1.8646980   0.2116066  -8.812 < 2e-16 ***
## InternetServiceFiber optic 1.1089214   0.1502548   7.380 1.58e-13 ***
## InternetServiceNo -0.9622809   0.1776636  -5.416 6.08e-08 ***
## tenure_group13-24 -0.2963519   0.1549252  -1.913   0.0558 .
## tenure_group25-48 -0.0853391   0.2719455  -0.314   0.7537
## tenure_group49-60  0.3995459   0.4348560   0.919   0.3582
## tenure_group61+    0.5080897   0.5402313   0.941   0.3470
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6113.6  on 5282  degrees of freedom
## Residual deviance: 4463.7  on 5271  degrees of freedom
## AIC: 4487.7
##
## Number of Fisher Scoring iterations: 6
```

```
exp(coef(glm_fit))
```

```
##              (Intercept)                tenure
##              0.8941692                0.9472102
##      MonthlyCharges                TotalCharges
##              1.0009620                1.0001601
##      ContractOne year                ContractTwo year
##              0.4338769                0.1549430
##      InternetServiceFiber optic                InternetServiceNo
##              3.0310872                0.3820206
##      tenure_group13-24                tenure_group25-48
##              0.7435257                0.9182009
##      tenure_group49-60                tenure_group61+
##              1.4911474                1.6621130
```

12. Model Evaluation

```
test_pred_prob <- predict(glm_fit, newdata = test_df, type = "response")

test_pred_class <- ifelse(test_pred_prob > 0.5, "Yes", "No") %>%
  factor(levels = levels(df$Churn))

confusionMatrix(test_pred_class, test_df$Churn, positive = "Yes")
```

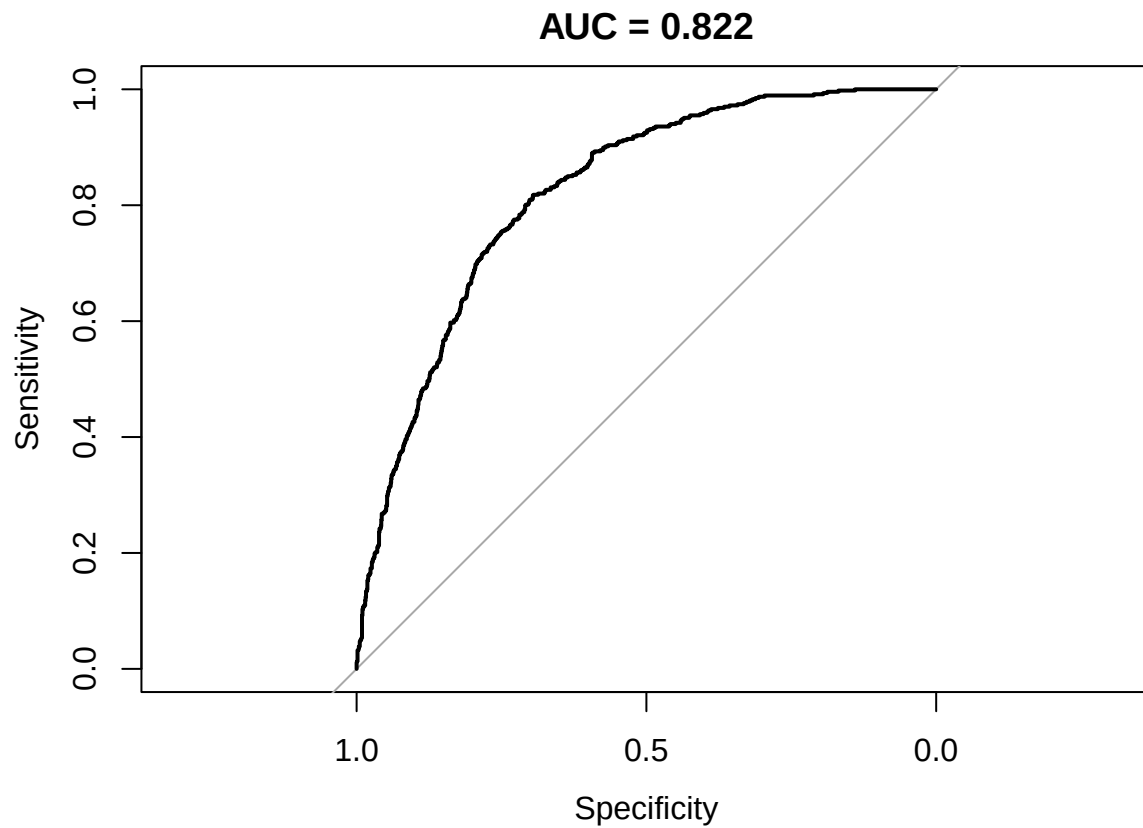
```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction   No  Yes
##      No  1181  282
##      Yes   112  185
##
##              Accuracy : 0.7761
##              95% CI : (0.7559, 0.7954)
##      No Information Rate : 0.7347
##      P-Value [Acc > NIR] : 3.441e-05
##
##              Kappa : 0.3503
##
##      McNemar's Test P-Value : < 2.2e-16
##
##              Sensitivity : 0.3961
##              Specificity : 0.9134
##      Pos Pred Value : 0.6229
##      Neg Pred Value : 0.8072
##      Prevalence : 0.2653
##      Detection Rate : 0.1051
##      Detection Prevalence : 0.1688
##      Balanced Accuracy : 0.6548
##
##      'Positive' Class : Yes
##
```

13. ROC Curve & AUC

```
roc_obj <- roc(  
  response = test_df$Churn,  
  predictor = test_pred_prob,  
  levels = c("No", "Yes")  
)  
  
auc(roc_obj)
```

Area under the curve: 0.8218

```
plot(roc_obj, main = paste("AUC =", round(auc(roc_obj), 3)))
```



14. Threshold Tuning

```
coords(  
  roc_obj,  
  x = "best",  
  best.method = "youden",  
  ret = c("threshold", "sensitivity", "specificity")  
)
```



```
## threshold sensitivity specificity
## 1 0.2418706 0.8179872 0.6952823
```

15. Variable Importance (GLM Approximation)

```
glm_summary <- summary(glm_fit)
zvals <- coef(glm_summary)[, "z value"]
sort(abs(zvals), decreasing = TRUE)
```

```
##          ContractTwo year InternetServiceFiber optic
##          8.8120957          7.3802709
##          ContractOne year          InternetServiceNo
##          6.8558919          5.4163077
##          tenure          tenure_group13-24
##          5.0447445          1.9128716
##          TotalCharges          tenure_group61+
##          1.8445189          0.9405039
##          tenure_group49-60          (Intercept)
##          0.9188003          0.4970491
##          tenure_group25-48          MonthlyCharges
##          0.3138094          0.2343162
```

16. Save Outputs

```
saveRDS(glm_fit, "glm_telco_model.rds")

write_csv(
  test_df %>%
    mutate(
      pred_prob = test_pred_prob,
      pred_class = test_pred_class
    ),
  "predictions_test.csv"
)
```